

PAPER_777: NGC 6217 — UQFF Barred Spiral Galaxy Standard Rotation

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Framework: UQFF (Universal Quantum Field Superconductive Framework)

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CP4 Class: #361 — NGC6217BarredSpiralUQFFCalculator

Abstract

NGC 6217 is a barred spiral galaxy ~67 million light-years distant ($z = 0.0045$) in the constellation Ursa Minor. It became notable as one of the first targets imaged by the Hubble Space Telescope's repaired Wide Field Camera 3 (WFC3) after the 2009 Servicing Mission 4, demonstrating the camera's restored capability. With moderate star formation ($\text{SFR} \approx 1 \text{ M}_\odot/\text{yr}$) and an extended rotating disk of $\sim 10^{11} \text{ M}_\odot$, NGC 6217 represents a typical SBbc barred spiral. Under UQFF, standard rotation velocity ($v = 10^5 \text{ m/s}$) and typical HII B-field yield $g_{\text{NGC6217}} \approx 1.053 \times 10^{-3} \text{ m/s}^2$.

1. Introduction

NGC 6217 (also catalogued as UGC 10470) has an apparent magnitude of ~ 11.2 , a disk diameter of $\sim 30,000 \text{ ly}$, and a Hubble morphological type of SBbc (barred spiral, late intermediate). Its bar structure funnels gas toward the central region, sustaining moderate star formation. The galaxy's selection as a Hubble Servicing Mission 4 first-light target in 2009 reflects its moderate brightness and interesting bar+ring morphology at $z = 0.0045$ ($\sim 67 \text{ Mly}$). UQFF treats it as a standard barred galaxy with SFR coupling through the galactic bar channel.

2. Master UQFF Gravity Equation

$$g_{\text{NGC6217}}(r, t) = (G \times M) / r^2 \times (1 + H(z) \times t) \times (1 + M_{\text{sf}}) \times (1 + f_{\text{TRZ}}) + a_{\text{EM}}$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	$10^{11} \text{ M}_\odot = 1.989 \times 10^{41} \text{ kg}$	Estimate
Disk radius	r	$3 \times 10^{20} \text{ m}$ ($\sim 30 \text{ kly}$)	NED
SFR	—	$1 \text{ M}_\odot/\text{yr}$	Moderate SBbc
Age (evolution time)	t	$5 \times 10^9 \text{ yr} = 1.578 \times 10^{17} \text{ s}$	Hubble time
M_sf	—	0.045	UQFF SFR integral
Redshift	z	0.0045	NED spectroscopic
v_EM	v	10^5 m/s	Disk rotation
B_EM	B	10^{-5} T	Galactic field
f_TRZ	—	0.04	UQFF

3. Long-Form Derivation

Step 1: Base Gravitational Term

$$\begin{aligned} g_{\text{grav}} &= G \times M / r^2 \\ &= 6.6743\text{e-}11 \times 1.989\text{e}41 / (3\text{e}20)^2 \\ &= 1.328\text{e}31 / 9\text{e}40 \\ &= 1.476\text{e-}10 \text{ m/s}^2 \end{aligned}$$

Step 2: Cosmic Expansion Factor

$$\begin{aligned} H(z) &= H_0 \times (1 + z)^{(3/2)} \approx 2.315\text{e-}18 \text{ s}^{-1} \\ H(z) \times t &= 2.315\text{e-}18 \times 1.578\text{e}17 = 0.3653 \\ 1 + H(z) \times t &= 1.3653 \end{aligned}$$

Step 3: Star-Formation Mass Fraction (Bar-Channeled)

$$\begin{aligned} \text{Bar-driven SFR} &= 1 \text{ M}_{\odot}/\text{yr}, \text{ integrated over } 5 \text{ Gyr} \\ M_{\text{sf}} &= 0.045 \text{ (UQFF bounded, } \sim 4.5\% \text{ mass fraction in young stars)} \\ 1 + M_{\text{sf}} &= 1.045 \end{aligned}$$

Step 4: Time-Reversal Correction

$$\begin{aligned} f_{\text{TRZ}} &= 0.04 \text{ (moderate spiral, bar-driven)} \\ 1 + f_{\text{TRZ}} &= 1.04 \end{aligned}$$

Step 5: Gravitational Total

$$\begin{aligned} g_{\text{grav_total}} &= 1.476\text{e-}10 \times 1.3653 \times 1.045 \times 1.04 \\ &= 1.476\text{e-}10 \times 1.488 = 2.197\text{e-}10 \text{ m/s}^2 \end{aligned}$$

Step 6: Aether Electromagnetic Correction

$$\begin{aligned} v &= 10^5 \text{ m/s (disk rotation velocity)} \\ B &= 10^{-5} \text{ T (galactic B-field)} \\ a_{\text{EM}} &= (e/m_p) \times (v \times B) \times \Lambda_{\text{UQFF}} \\ &= 9.575\text{e}7 \times (10^5 \times 10^{-5}) \times 11 \times 10^{-12} \\ &= 9.575\text{e}7 \times 1 \times 1.1\text{e-}11 \\ &= 1.053\text{e-}3 \text{ m/s}^2 \end{aligned}$$

Step 7: Final Solution

$$\begin{aligned} g_{\text{NGC6217}} &= 2.197\text{e-}10 + 1.053\text{e-}3 \\ &\approx 1.053\text{e-}3 \text{ m/s}^2 \end{aligned}$$

4. Physical Interpretation

The classical gravitational term ($2.197 \times 10^{-10} \text{ m/s}^2$) is 7 orders of magnitude smaller than the Aether EM result ($1.053 \times 10^{-3} \text{ m/s}^2$). The UQFF result thus captures disk rotation dynamics through the electromagnetic Aether coupling, not Newtonian gravity. The bar structure in NGC 6217 channels gas inward, sustaining the moderate $\text{SFR} = 1 \text{ M}_{\odot}/\text{yr}$ and bar-enhanced $M_{\text{sf}} = 0.045$, slightly higher than a purely symmetric flocculent spiral

of similar mass. As with NGC 2841, the dominant physics is electromagnetic at these rotation velocities.

5. UQFF Framework Advancement

- NGC 6217 validates standard SBbc barred spiral UQFF field result
 - Bar-driven $\text{SFR} = 1 \text{ M}_\odot/\text{yr}$ produces $M_{\text{sf}} = 0.045$ (canonical bar value)
 - Hubble SM4 first-light observation history preserved in UQFF canon
 - SBbc result consistent with other moderate barred spirals (NGC 2841, NGC 1300)
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6. Conclusions

UQFF applied to NGC 6217 yields $g_{\text{bar spiral}} \approx 1.053 \times 10^{-3} \text{ m/s}^2$, confirming standard barred galaxy behavior. The Hubble-celebrated target joins the UQFF canon as the SBbc benchmark alongside Milky Way analogs. Bar-enhanced gas flow, expressed through $M_{\text{sf}} = 0.045$ and moderate SFR, demonstrates how UQFF differentiates galaxy morphological types within a unified framework.

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